

Facial Animation at Scale

Producing facial animation for 24 hours of narrative content in “Horizon: Forbidden West”

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ABSTRACT

This talk covers the complete pipeline and process used to generate facial animation data for all the narrative content in the latest installment of the Horizon game series, created by Guerrilla. We developed a proprietary technique for embedding per-frame timecode and source take information into all our narrative animation data as the basis of this pipeline. Then we added a fully automated facial animation solving pipeline where animators could push animation quality by improving generic FaceWare models. Any update to these models would automatically cause the re-processing of all dependent facial animation data. To manage all this data a review pipeline was set up in Atlassian Jira that provided both a high-level overview as well as granular control over the more than 15,000 narrative facial animations in the game.

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1 INTRODUCTION

One of the goals when starting on the development of Horizon Forbidden West was to make sure that the quality of the facial animation would be consistent throughout the entire game. In the first installment of the Horizon series (Horizon Zero Dawn) we were able to deliver decent quality facial animation in the important cutscenes driving the main storyline. However, once the player interacted with NPC characters in one of the many side quests the game relied on procedurally generated animation of much lower quality. This resulted in a noticeable contrast in quality and degraded the suspension of disbelief. The first game shipped with roughly 3 hours of high-quality cinematics and about 7 hours of secondary NPC conversation content. Once complete, Horizon Forbidden West shipped with over 8 hours of cinematic content and over 16 hours of NPC conversations. This meant that we had to develop a pipeline for producing high quality facial animation for approximately 24 hours of narrative content, compared to the 3 hours of the previous game.

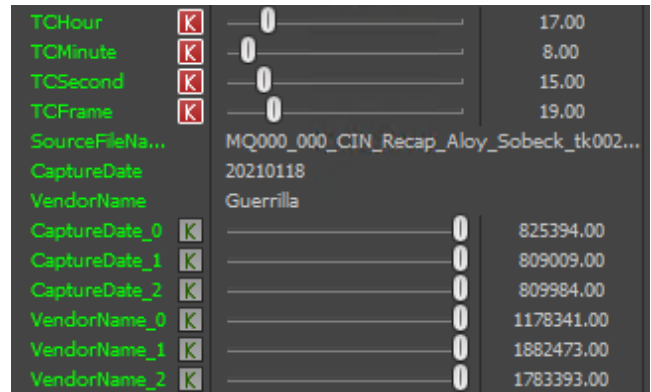
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Figure 1: Encoded take data stored as animatable integers.

2 ENCODED TIMECODE

Before we could solve the problem of scope, we had to find a method for maintaining sync. When recording Performance Capture (capturing body motion & face video at the same time) the recorded data is obviously in sync. But, in the following phases of the cinematic animation workflow the recorded body motion is heavily manipulated, to the point where frames from multiple takes can be stitched together to create a single animation. The problem we had to solve was how to make sure that at any point in the pipeline we can sync frames from the face video to the matching frames of the body data. The actual timecode time data itself is trivial to store per frame as it’s represented by numeric data that’s easily animated over time. However, to sync body data to the correct face video we also needed to know which source video the timecode time belonged to. This was solved by encoding the data of the take name, session date and mocap vendor into a set of animatable integers and storing that data on every frame of the original motion capture data.

All this time code data was generated as soon as motion capture data was delivered and would stay with the animation data all the way through the pipeline and even stored in our proprietary game engine animation format. This talk will illustrate examples of how this data was used in various softwares and phases of production, as well as highlight some of the unexpected issues we had to troubleshoot along the way.

3 SOLVING THE FACES

The core of the facial animation solving pipeline for Horizon Forbidden West was the tracking and solving software developed by Faceware Technologies. Both pieces of software rely on using pre-build models to assist the automated processes. Our solving pipeline was

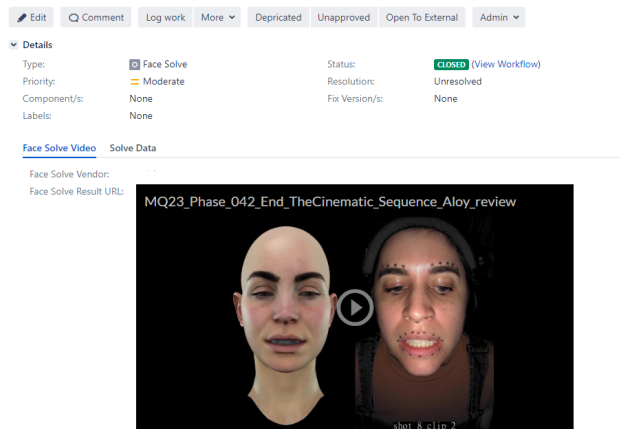


Figure 2: The Jira page for a single solved HMC video

based on the concept of having our facial animators update these models and then having an automated system determine which animations need re-processing based on the modified models. We implemented a tier-based system to manage various levels of control in these models. At the lowest level we had a generic tracking and solving model, the next level would be an actor-based model tailored to a specific actor's face, followed by a motion capture session-based mode for a specific actor and finally a take specific model for the actor. This way when new footage came in we would immediately be able to process it with the generic models, and then start refining them by creating or improving more granular models based on the specifics of the footage. In practice there were some

sessions that delivered decent results with more generic models while some required dedicated attention and individual models.

By basing our pipeline on improving the underlying models instead of the individual animations we were able to make sure all facial animations were of a consistent quality and reduced the chance of having content fall through the cracks as the volume increased.

4 KEEPING TRACK

One of the major problems with the volume of facial animation on the previous game Horizon Zero Dawn was that of quality assurance. There were no ways for anyone to easily view and check all the solved animations in a manageable amount of time. Knowing that the volume of content would increase significantly it was critical that we maintained the ability to efficiently view the results of the solves. Since we were already using Atlassian Jira for tracking most of our production we decided to extend that and also use it to track the facial solving content. Each character's performance for a game sequence was represented in a Jira Ticket; here we stored meta-data about the scene like the actor(s) involved, the session name and a time code track for the required HMC videos. Once such a sequence had been solved, we also attached a side-by-side video showing the original HMC video next to a rendered video of the solved animation. By changing the status on these Jira tickets, we could further control which sequences would be picked up for re-processing or which ones were locked for further animation polish. The real benefit of this Jira based system became apparent when we were able to involve our internal QA team to validate all the videos for tracking/solving errors and mark those tickets. We could now easily identify which animations needed additional time spent on their tracking and/or solving models.